

Experiment No. 2

Title: Study of IC-74LS83 as a BCD adder.

Problem Definition & Aim of Experiment:

1. Design(truth table, K-map) and Implement 1 digit BCD adder using IC-74LS83
- 2.

Objective of Experiment:

To understand how both arithmetic operations, addition in BCD can be performed by using IC 7483.

Lab Equipment's and IC's Used:

Digital Trainers Kit 74LS83, 74LS08, 74LS32, 74LS04.

Theory:

BCD Adder:

BCD adder is a circuit that adds two BCD digits & produces a sum of digits also in BCD.

Rules for BCD addition:

1. Add two numbers using rules of Binary addition.
2. If the 4 bit sum is greater than 9 or if carry is generated then the sum is invalid. To correct the sum add 0110 i. e. (6)10 to sum. If carry is generated from this addition add it to next higher order BCD digit.
3. If the 4 bit sum is less than 9 or equal to 9 then sum is in proper form.

CASE I : Sum ≤ 9 & carry = 0.

Add BCD digits 3 & 4

$$\begin{array}{r} 1. \quad 0011 \\ + \quad 0100 \\ \hline 0111 \end{array}$$

Answer is valid BCD number = (7)BCD & so 0110 is not added.

CASE II : Sum > 9 & carry = 0.

Add BCD digits 6 & 5

$$\begin{array}{r} 1. \quad 0110 \\ + \quad 0101 \\ \hline 1011 \end{array}$$

Invalid BCD (since sum > 9) so 0110 is to be added

$$\begin{array}{r} 2. \quad 1011 \\ + \quad 0110 \\ \hline 1 \quad 0001 \end{array}$$

(1 1)BCD

Valid BCD result = (11) BCD

CASE III : Sum <= 9 & carry = 1.

Add BCD digits 9 & 9

1. 1 0 0 1

+ 1 0 0 1

1 0 0 1 0

Invalid BCD (since Carry = 1) so 0110 is to be added

2. 1 0 0 1 0

+ 0 1 1 0

1 1 0 0 0

(1 8)BCD

Valid BCD result = (18) BCD

Design of BCD adder :

1. To execute first step i. e. binary addition of two 4 bit numbers we will use IC 7483
(with Cin = 0), which is 4 bit binary adder.
2. We need to design a digital circuit which will sense sum & carry of IC 7483 & if sum exceeds 9 or carry = 1, this digital circuit will produce high output otherwise its output will be zero.

Circuit to check invalid BCD :

First we will design circuit to check sum & then we will logically OR output of this circuit to carry output of IC 7483

For digital circuit which we are going to design we will have 4 inputs(S3, S2, S1, S0) & only 1 output Y.

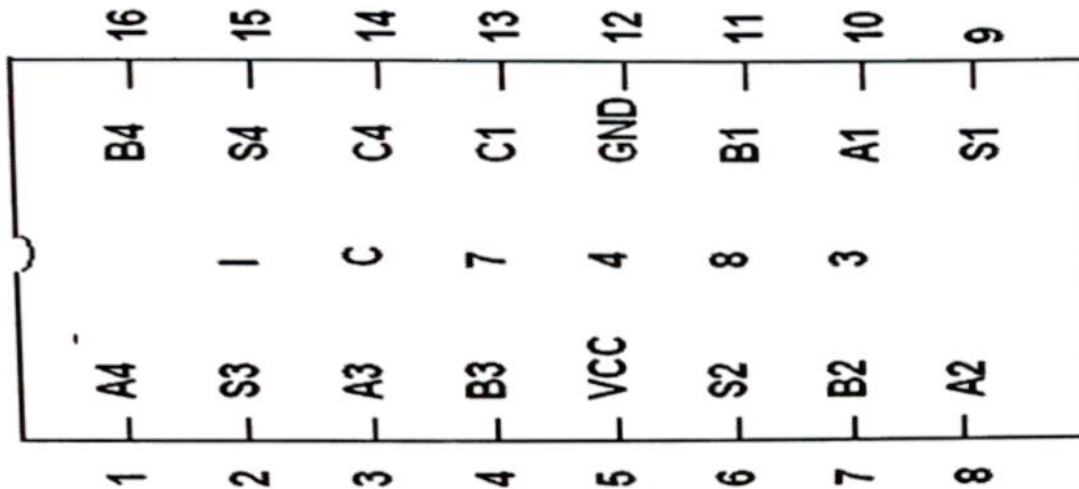
a) Y output of this circuit. Will be ORed with carry output of first adder IC 7483.

b) If BCD result is invalid i. e. sum output of first 7483 we have to add (6)₁₀ i.e. (0110)₂ that means we need one more binary adder IC 7483.

c) If BCD result is valid (i.e. final output of the circuit to check validity is 0) we will make an arrangement that second adder IC 7483 adds (0)₁₀ i. e. (0000)₂to

the sum of the first adder IC 7483. The output of the combinational circuit is used as final output carry & carry output of second adder IC is ignored.

Pin Diagram of IC7483:



TruthTable for invalid BCD numbers:

	Inputs				Output	
	S_3	S_2	S_1	S_0	Y	
0	0	0	0	0	0	Sum is valid BCD number $\therefore Y=0$
1	0	0	0	1	0	
2	0	0	1	0	0	
3	0	0	1	1	0	
4	0	1	0	0	0	
5	0	1	0	1	0	
6	0	1	1	0	0	
7	0	1	1	1	0	
8	1	0	0	0	0	Sum is invalid BCD number $\therefore Y=1$
9	1	0	0	1	0	
10	1	0	1	0	1	
11	1	0	1	1	1	
12	1	1	0	0	1	
13	1	1	0	1	1	
14	1	1	1	0	1	
15	1	1	1	1	1	

Write Kmap:-

		$S_3 S_2$			
$S_1 S_0$	00	01	11	10	
00	0	0	1	0	
01	1	0	0	0	
11	0	0	1	1	
10	0	0	1	1	

$$Y = S_3 S_2 + S_3 S_1$$

$$Y = S_3 (S_2 + S_1)$$

K Map for Output Y:

- i) The o/p of combinational ckt should be 1 if Carry of adder-1 is high
 $\therefore Y$ is ORed with Carry of adder 1 as shown in fig.
- ii) The O/p of combinational ckt is connected to B_1, B_2 i/p's of adder-2 & $B_3 = B_0 = 0$ as they are connected to ground permanently.
 This makes $B_3 B_2 B_1 B_0 = 0110$ if $Y' = 1$
- iii) The sum o/p's of adder-1 are applied to $A_3 A_2 A_1 A_0$ of adder-2.

BCD ADDER DIAGRAM:

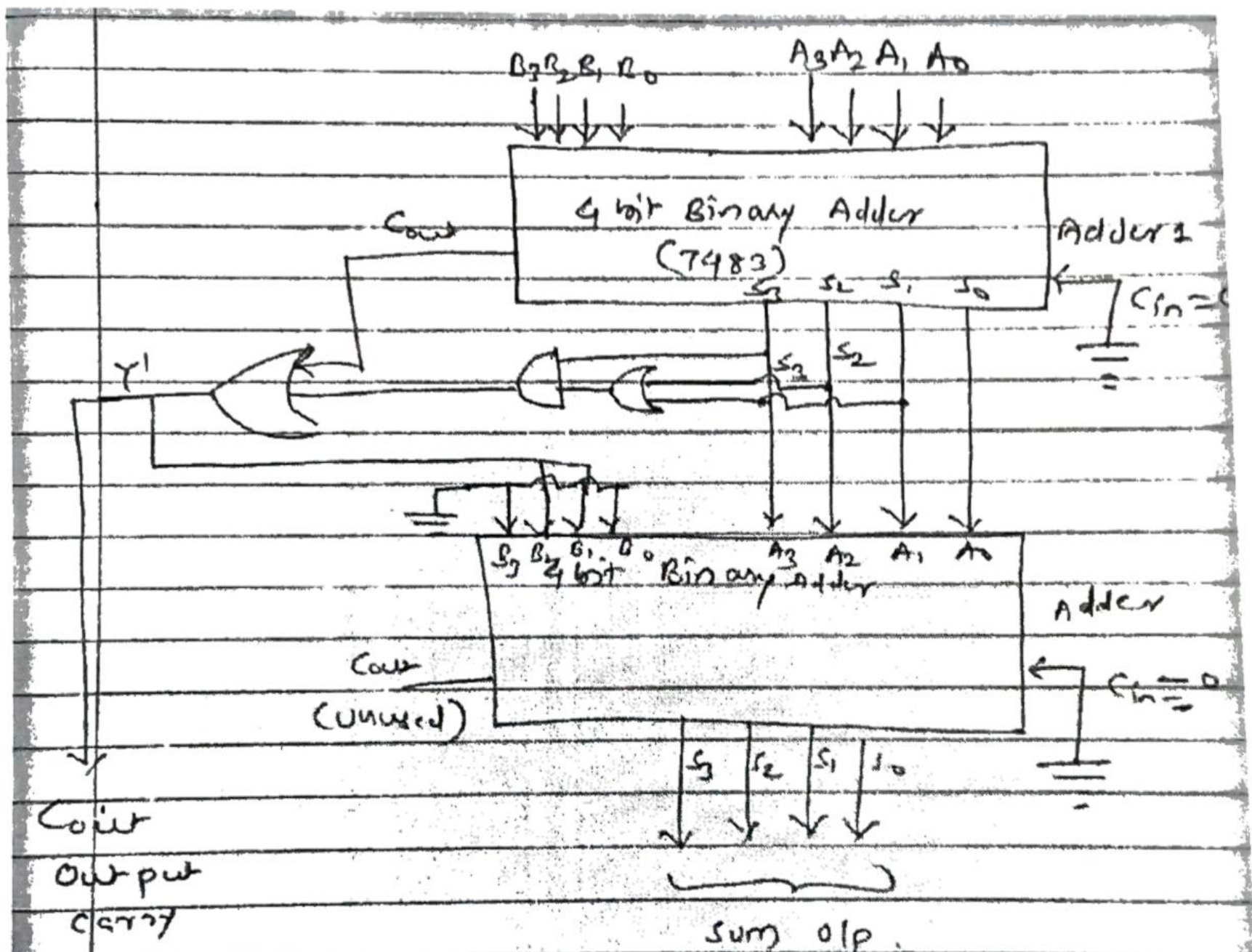


fig. 4-bit BCD Addition.